

Explaining Visual Illusions Through Perception

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Senior Secondary (Years 11-12)

~30 minutes

Science · Psychology

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Simulation: <https://michaelbach.de/ot/>

Curriculum links: perception · sensation · brain function

Learning intentions

- can distinguish between sensation and perception using real visual illusion examples.
- can explain the perceptual mechanism behind at least three different illusions.
- can evaluate why the same visual input can be perceived differently by different viewers.

Getting started

Your teacher will direct you to a small set of illusions from the collection relevant to today's topic. For each one, view the illusion first without reading the explanation, record your own perception, then read the built-in explanation before answering.

Tasks

1. For your first assigned illusion, describe exactly what you perceive before reading any explanation.

2. Now read the site's explanation for that illusion. In one sentence, summarise the perceptual mechanism it describes.

3. Using the terms 'sensation' and 'perception', explain the difference between what your eyes physically detected in this illusion and what your brain constructed as your experience of it.

4. Repeat this process for three illusions total, recording the mechanism behind each in your own words.

Illusion name	Perceptual mechanism (your own words)

5. Choose one illusion from your table that relies on colour or brightness perception. Which part of the visual system (e.g. retina, visual cortex) is most likely involved, based on the explanation given?

6. Some illusions work on almost everyone, while others depend on individual differences (e.g. colour vision, prior experience). Choose one illusion and evaluate whether it seems to be a universal effect or one that could vary between viewers.

7. Explain why optical illusions are useful evidence for psychologists studying perception, even though they involve 'incorrect' perceptions rather than accurate ones.

Extension (optional)

Choose one illusion not covered in your table and design a simple real-world scenario (e.g. road design, packaging, user interface) where this specific perceptual effect could cause a genuine practical problem if not accounted for.